

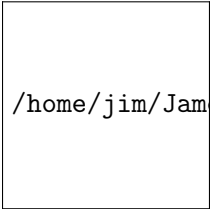
Section 3.5: Expected Value

Finite Mathematics MTH 107 Fall 2014

Dr. James Ashe

Mars Hill University

Friday 17th October, 2014



`/home/jim/JamesAshe/MHU/images/MarsHillU`

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John places bets of \$70, \$80, \$50, and \$100 on a game in which he has a 25% chance of winning. How much money should he expect to win?

$$70 \cdot \frac{1}{4} + 80 \cdot \frac{1}{4} + 50 \cdot \frac{1}{4} + 100 \cdot \frac{1}{4} = \$75$$

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- A summary of what *has* happened.

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- A summary of what *will* happen.

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Example 3.5.1.

A saleswoman has found that her weekly commissions have the following probabilities.

Expected Commission:

$$= (0)(.05)$$

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A saleswoman has found that her weekly commissions have the following probabilities.

commision	0	\$100	\$200	\$300
probability	0.05	0.35	0.50	0.10

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$$= (0)(.05) + (100)(.35)$$

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$$= (0)(.05) + (100)(.35) + (200)(.50) + (300)(.10) = \$165$$

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$$= (0)(.05) + (100)(.35) + (200)(.50) + (300)(.10) = \$165$$

- Based on her sales history, she should expect to make \$165 a week in commissions.

Example 3.5.2.

You are asked to play the following game:

Should you play?

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You are asked to play the following game:

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$$p(1 \cup 2) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

$$p(3) = \frac{1}{6}$$

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$$EV = 50 \cdot \frac{1}{3}$$

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$$EV = 50 \cdot \frac{1}{3} - 20 \cdot \frac{1}{6}$$

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So NO.

Example 3.5.3.

A club is having a raffle. Tickets cost \$5, but each prize will be given away regardless of tickets sold. Should you buy a ticket under the following circumstances:

a. 1000 tickets are sold.

- Subtract \$5 from the value of the prizes for the ticket price.

a. $EV =$

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A club is having a raffle. Tickets cost \$5, but each prize will be given away regardless of tickets sold. Should you buy a ticket under the following circumstances:

- a. 1000 tickets are sold.

Prize	Value	#
Laptop	\$2,325	1
MP3 Player	\$425	2
20 CDs	\$320	3
A Pizza	\$23	4
Book	\$20	15

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a. $EV = 2320\left(\frac{1}{1000}\right)$

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- There are $1000 - 25 = 975$ losing tickets.

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 $\approx -\$0.473$

Example 3.5.3.

A club is having a raffle. Tickets cost \$5, but each prize will be given away regardless of tickets sold. Should you buy a ticket under the following circumstances:

- a. 1000 tickets are sold.
- b. 700 tickets are sold.

Prize	Value	#
Laptop	\$2,325	1
MP3 Player	\$425	2
20 CDs	\$320	3
A Pizza	\$23	4
Book	\$20	15
nothing	\$0	

- Subtract \$5 from the value of the prizes for the ticket price.
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b. $EV = 2320\left(\frac{1}{700}\right)$
 $+ 420\left(\frac{2}{700}\right) + 315\left(\frac{3}{700}\right) + 18\left(\frac{4}{700}\right) + 15\left(\frac{15}{700}\right)$

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Book	\$20	15
nothing	\$0	675

- Subtract \$5 from the value of the prizes for the ticket price.
- There are $700 - 25 = 675$ losing tickets.

a. $EV = 2320\left(\frac{1}{1000}\right) + 420\left(\frac{2}{1000}\right) + 315\left(\frac{3}{1000}\right) + 18\left(\frac{4}{1000}\right) + 15\left(\frac{15}{1000}\right) - 5\left(\frac{975}{1000}\right)$
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b. $EV = 2320\left(\frac{1}{700}\right)$
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b. $EV = 2320\left(\frac{1}{700}\right)$
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